PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important).

- *Access to good healthcare is still a serious challenge for many people, especially those living in semi-urban and rural areas. Patients often struggle with long queues, uncertainty about doctor availability, and the hassle of booking appointments. In many clinics, records are still maintained manually, which increases the chances of errors, misplaced files, and inefficient patient management.*
- *The situation becomes even more difficult during emergencies or pandemics, when visiting hospitals in person can be unsafe or simply not possible. Patients need reliable ways to consult doctors remotely, track their medicines, receive health updates, and monitor their conditions without always traveling to a hospital.*
- *To address these challenges, we propose **Arogyam**, a smart healthcare web application designed to digitally connect patients and doctors on a single platform. Arogyam supports both online and offline appointment booking, along with video consultations and chat-based communication. Patients can easily reschedule appointments, track prescribed medicines, view health notes shared by doctors, and stay informed about current medical issues and safety precautions.*
- *Our vision is to build a centralized, user-friendly system that makes healthcare more accessible, reduces waiting time, improves doctor–patient communication, and securely maintains digital health records.*
- *Arogyam ultimately aims to simplify healthcare management and ensure that quality medical services are available to everyone in a more organized, efficient, and accessible way.*

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is)).

- *At present, many hospitals and clinics depend on basic appointment booking systems or third-party platforms for managing patient interactions. Popular applications such as Practo and hospital-specific portals offer online booking and consultation services. However, a large number of small and local clinics still rely on manual registers or phone calls to schedule appointments.*
- *Most existing systems also come with several limitations. They usually do not combine essential features like chat, video consultation, medicine tracking, and appointment booking into a single platform. Real-time appointment rescheduling is often missing, and both doctors and patients rarely get personalized dashboards to manage their activities efficiently. In addition, tracking past appointments, prescriptions, and health updates is usually limited or poorly organized.*
- *Another major concern is cost. Many small clinics cannot afford complex healthcare management systems because of their high setup and maintenance expenses.*
- *To overcome these challenges, **Arogyam** is designed as a cost-effective, fully integrated healthcare solution built using the MERN stack. It brings together online and offline appointment booking, video consultation, chat communication, medicine tracking, dedicated dashboards for doctors and patients, and health content publishing—everything in one simple and powerful system.*

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones.).

General Goals:

- *Develop a full-stack healthcare management web application.*
- *Enable online and offline appointment booking.*
- *Implement real-time chat and video consultation.*
- *Provide medicine tracking and booking history.*
- *Allow doctors to publish health notes and case studies.*
- *Provide secure authentication and data management.*

Initial Milestones:

1. *Requirement Analysis and System Design*
2. *Database Schema Design (Patients, Doctors, Appointments, Chat, Call Schedules)*
3. *Backend API Development using Node.js & Express*
4. *Frontend UI Development using React*

Secondary Milestones:

5. *Appointment Booking and Payment Integration*
6. *Video Call & Chat System Integration*
7. *Medicine Tracking and Health Notes Module*
8. *Role-Based Authentication (Admin, Doctor, Patient)*

Final Milestones:

9. *Testing and Bug Fixing*
10. *Deployment on Cloud*
11. *Performance Optimization*
12. *Documentation and Presentation*

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use).

*We are using the **MERN Stack** (MongoDB, Express.js, React.js, Node.js) to build Arogyam.*

Backend:

- *Node.js and Express.js for server-side logic.*
- *MongoDB for database management.*
- *REST APIs for communication between frontend and backend.*
- *JWT for authentication and authorization.*
- *Role-based access control (Admin, Doctor, Patient).*

Frontend:

- *React.js for dynamic and responsive UI.*
- *Component-based structure.*
- *Context API for state management.*
- *Axios for API communication.*

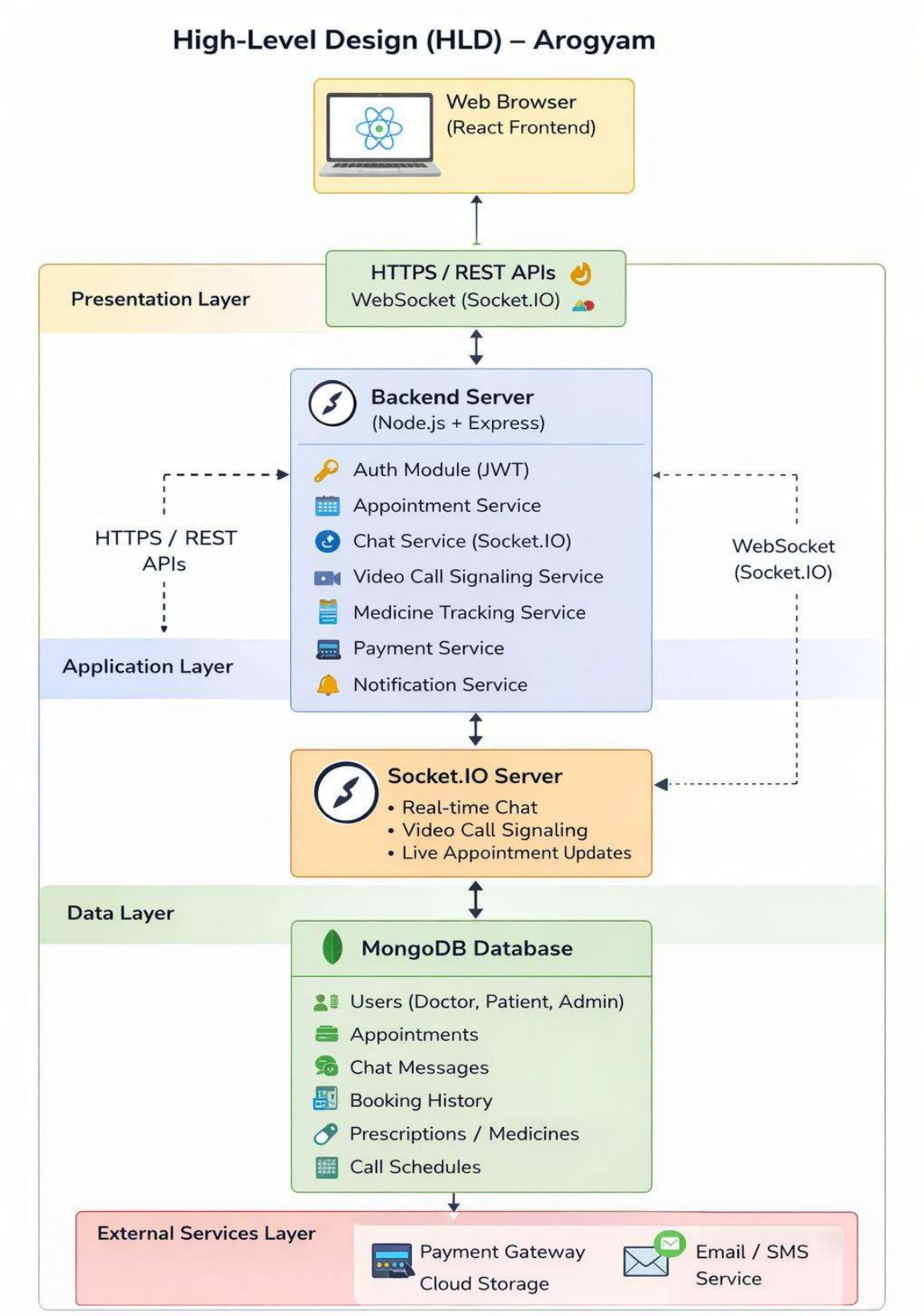
Key Features Implementation:

- *Appointment scheduling system with status tracking.*
- *Video calling system for doctor-patient consultation.*
- *Real-time chat system.*
- *Medicine and prescription tracking.*
- *Booking history for doctors and patients.*
- *Dashboard for analytics and management.*

The system is designed in a modular architecture with separate folders for Models, Controllers, Routes, and Middleware, ensuring maintainability and scalability.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project).

The final deliverables of Arogyam include:

- *Fully functional MERN-stack web application.*
- *Patient dashboard with booking, chat, video call, medicine tracking.*
- *Doctor dashboard with appointment management, case study publication, notes, and chat.*
- *Admin panel for system monitoring.*
- *Secure authentication and role-based access.*
- *Appointment history and booking management.*
- *Integrated online payment system.*
- *Real-time chat and video consultation feature.*
- *Project documentation and source code.*
- *Deployment-ready application.*

The system will provide an efficient digital healthcare solution for clinics and patients.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem.)

- *Users have internet access.*
- *Doctors and patients will register with valid credentials.*
- *Basic knowledge of using web applications.*
- *Video call performance depends on internet quality.*
- *MongoDB cloud server remains available.*
- *Payment gateway integration assumes valid merchant setup.*

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

- *MERN Stack Documentation*
  https://www.mongodb.com
  https://expressjs.com
  https://reactjs.org
  https://nodejs.org
- *JWT Authentication*
  https://jwt.io
- *REST API Design Principles*
  https://restfulapi.net
- *Healthcare Management Research Articles (Google Scholar)*